XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	COM106			
Course Name:	Electrical Engineering			
Question Paper Setter:	Assoc. Prof. Ir. Ts. Dr. Wong Shen Yuong			
Academic Session:	2022/04	Question Paper:	A☑	B ☐
Total No. of Pages:	5	Time Allocated:	2 hours	
Additional Materials:	-			
Apparatus Allowed:	Calculator			

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 5 Questions.

   Answer **all** questions.
2. Your answers must be handwritten.
3. Please follow the requirement of each question and paste on the downloadable answer book provided.
4. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

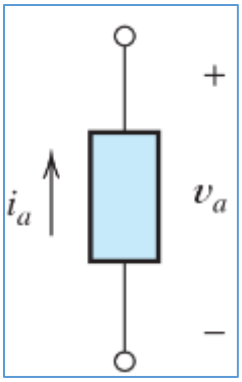
(Student ID: _____ Full Name: _____)

Question 1 [20 marks]

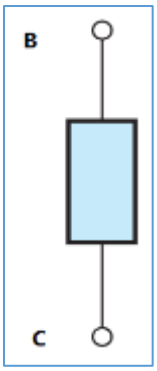
(CLO 1, PLO 1, C3)

(a) Suppose that the net charge passing through the cross section of a certain circuit element is given by $q(t) = 2t + 3t^2C$, with time *t* in seconds. Determine the current through the element as a function of time. [5 marks]

(b) Compute the power for each element shown in Fig. 1. For each case, state whether energy is being absorbed by the element or supplied by it. [10 marks]



(i)



(ii)

Figure 1

For case(i): $va = 15V, ia = -3A$

For case(ii): $v_{CB} = 20V, i_{BC} = 2A$

(c) Using current division principle, solve *Ix* in Fig. 2. [5 marks]

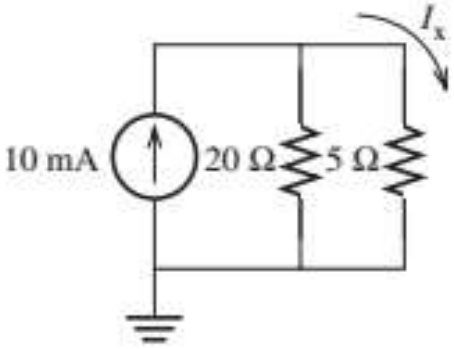


Figure 2

Question 2 [14 marks]

(CLO 1, PLO 1, C3)

Referring to Fig 3, apply mesh analysis to find currents i_1, and i_2 in the circuit.

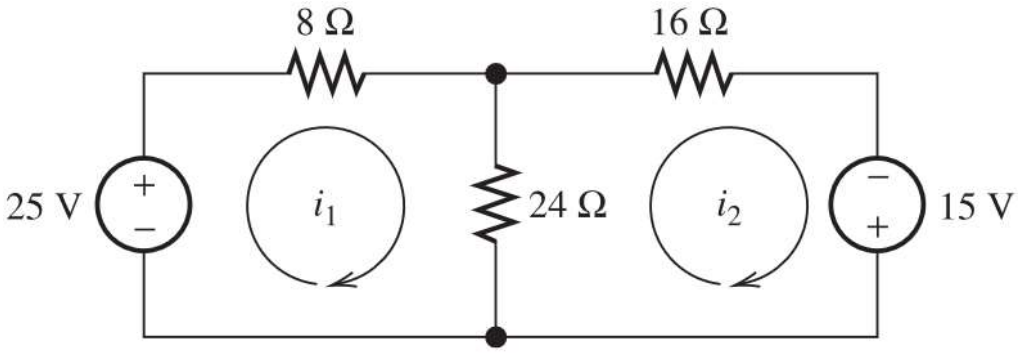


Figure 3

Question 3 [24 marks]

(CLO 2, PLO 2, C4)

Referring to Fig. 4,

(a) Find the Thevenin and Norton equivalent. [18 marks]

(b) Refer to your answer in (a), calculate the maximum power that can be delivered to the load. [4 marks]

(c) Refer to your answer in (b), determine what value of the load resistance is the power maximum? [2 marks]

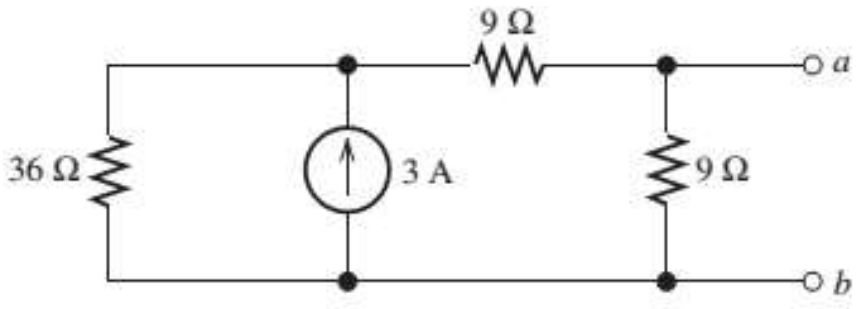


Figure 4

Question 4 [20 marks]

(CLO 2, PLO 2, C3)

(a) Using **source transformation**, solve current i_1 and i_2 in Fig. 5. [8 marks]

Show the diagram when solving for the current.

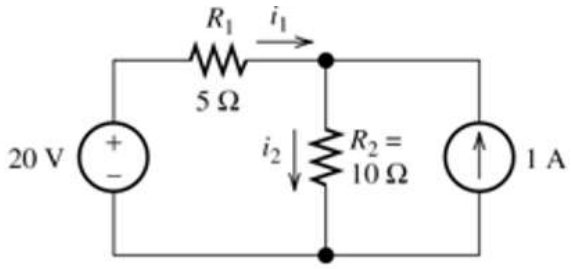


Figure 5

(b) Referring to Fig. 6, using the principle of **superposition**, find the current ***i***. [12 marks]

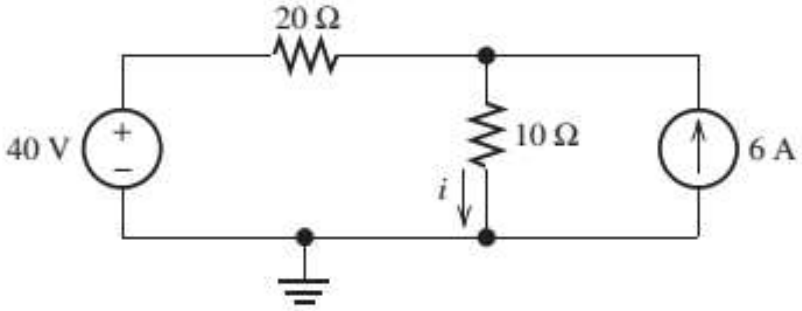


Figure 6

Question 5 [22 marks]

(CLO 2, PLO 2, C3)

Referring to Fig. 7,

(a) Produce the phasors for the current ***I*** and for the voltages ***Vs, V_R, Vc*** of the circuit. [12 marks]

(b) Present a phasor diagram showing ***Vs, V_R, Vc***, current ***I***. [8 marks]

(c) Assess the relationship between ***V_R,*** and ***V_C***. [2 marks]

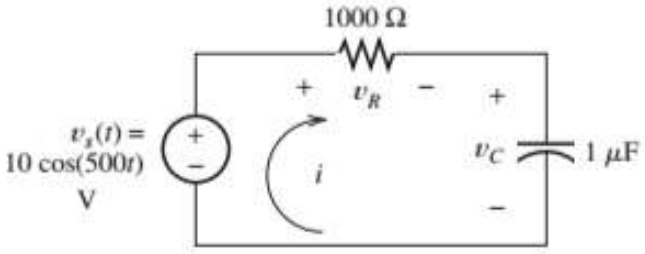


Figure 7

- **END OF QUESTIONS -**